Assignment 4: Normalization

Exercise 1: Happy Supplies Parts Order Form

a. Assumptions

1. Each form is one order. There is no order number and none may be added, so an order is identified by customerNumber + orderDate + orderTime.
2. customerNumber identifies a customer, and each customer has one name and one type. Names are not unique.
3. The employee depends on the order, not the customer, since employees serve any customer.
4. Each part is stored in exactly one cage (cages are not assigned by part type: a cage can hold several types, and one type can be stored in several cages) and has a fixed list price, the same on every order.
5. A part appears at most once per order; quantityOrdered covers multiples.

b. Step 0: all the data in one table (not in any normal form)

order_form: the form's data as written, with labels renamed to camelCase (Date → orderDate, Time → orderTime, Name → partName, Type → partType, Employee → employeeName).

customerNumber	orderDate	orderTime	customerName	customerType	employeeName	partNumber	partName	partType	cageCode	quantityOrdered	unitPrice
HG54587	7/1/2024	10:30am	Jeff Peterson	Consumer	D. Harrison	10654	Float Control	Plumbing	G413	4	12
						10456	Modulator	Electrical	H433	3	7
						10776	Hose Assembly	Plumbing	G413	7	9
						10657	Float Assembly	Plumbing	G413	5	10

Normal form: **none (unnormalized)**. PK: customerNumber, orderDate, orderTime.

- partNumber, partName, partType, cageCode, quantityOrdered and unitPrice form a **repeating group**: one order key value carries four sets of part values.
- Rows 2–4 have no PK value, which breaks entity integrity.

Step 1: first normal form (1NF)

order_form: one row per part on the order, every cell filled.

customerNumber	orderDate	orderTime	partNumber	customerName	customerType	employeeName	partName	partType	cageCode	quantityOrdered	unitPrice
HG54587	7/1/2024	10:30am	10654	Jeff Peterson	Consumer	D. Harrison	Float Control	Plumbing	G413	4	12
HG54587	7/1/2024	10:30am	10456	Jeff Peterson	Consumer	D. Harrison	Modulator	Electrical	H433	3	7
HG54587	7/1/2024	10:30am	10776	Jeff Peterson	Consumer	D. Harrison	Hose Assembly	Plumbing	G413	7	9
HG54587	7/1/2024	10:30am	10657	Jeff Peterson	Consumer	D. Harrison	Float Assembly	Plumbing	G413	5	10

Normal form: **1NF**. PK: customerNumber, orderDate, orderTime, partNumber.

Why each piece changed:

- **customerNumber, orderDate, orderTime, customerName, customerType, employeeName**: copied into rows 2–4, so every cell holds one value and the repeating group is gone.
- **partNumber**: added to the PK, since the order key now repeats on four rows. A part appears once per order (assumption 5), so the four attributes together are unique.

The table is not in 2NF. Its dependencies:

- customerNumber → customerName, customerType (**partial**: depends on one part of the PK)
- customerNumber, orderDate, orderTime → employeeName (**partial**: depends on the order, not the part)
- partNumber → partName, partType, cageCode, unitPrice (**partial**)
- customerNumber, orderDate, orderTime, partNumber → quantityOrdered (**full**)

Step 2: second normal form (2NF)

Each partial dependency moves to its own relation, keyed by the part of the PK it depends on.

customer

customerNumber	customerName	customerType
HG54587	Jeff Peterson	Consumer

customer_order

customerNumber	orderDate	orderTime	employeeName
HG54587	7/1/2024	10:30am	D. Harrison

part

partNumber	partName	partType	cageCode	unitPrice
10654	Float Control	Plumbing	G413	12
10456	Modulator	Electrical	H433	7
10776	Hose Assembly	Plumbing	G413	9
10657	Float Assembly	Plumbing	G413	10

order_line

customerNumber	orderDate	orderTime	partNumber	quantityOrdered
HG54587	7/1/2024	10:30am	10654	4
HG54587	7/1/2024	10:30am	10456	3
HG54587	7/1/2024	10:30am	10776	7
HG54587	7/1/2024	10:30am	10657	5

Why each piece changed:

- **customerName, customerType**: moved to **customer**, keyed by customerNumber, because they depend only on customerNumber.
- **employeeName**: moved to **customer_order**, keyed by customerNumber, orderDate, orderTime, because it depends on the order and not on partNumber.

- **partName, partType, cageCode, unitPrice**: moved to **part**, keyed by partNumber, because they depend only on partNumber.
- **quantityOrdered**: stays with the full four-attribute PK in the remaining relation, named **order_line**, because it depends on both the order and the part.

All four relations are now in 2NF: every non-key attribute depends on its relation's whole PK.

Step 3: third normal form (3NF)

No transitive dependencies exist, so no attribute moves and the 2NF relations are already in 3NF:

- **customer**: customerName does not determine customerType, because names are not unique (assumption 2).
- **customer_order**: employeeName is the only non-key attribute, so no non-key attribute can depend on another.
- **part**: cageCode does not determine partType, and partType does not determine cageCode (assumption 4).
- **order_line**: quantityOrdered is the only non-key attribute.

The FKs that link the relations are italicized below.

customer

customerNumber	customerName	customerType
HG54587	Jeff Peterson	Consumer

customer_order

customerNumber	orderDate	orderTime	employeeName
HG54587	7/1/2024	10:30am	D. Harrison

part

partNumber	partName	partType	cageCode	unitPrice
10654	Float Control	Plumbing	G413	12

partNumber	partName	partType	cageCode	unitPrice
10456	Modulator	Electrical	H433	7
10776	Hose Assembly	Plumbing	G413	9
10657	Float Assembly	Plumbing	G413	10

order_line

customerNumber	***orderDate***	***orderTime***	***partNumber***	quantityOrdered
HG54587	7/1/2024	10:30am	10654	4
HG54587	7/1/2024	10:30am	10456	3
HG54587	7/1/2024	10:30am	10776	7
HG54587	7/1/2024	10:30am	10657	5

c/d. Final 3NF relations: keys

Relation	PK	FK
customer	customerNumber	none
customer_order	customerNumber, orderDate, orderTime	customerNumber → customer.customerNumber
part	partNumber	none
order_line	customerNumber, orderDate, orderTime, partNumber	(customerNumber, orderDate, orderTime) → customer_order; partNumber → part.partNumber

Exercise 2: Panacea Mental Health Therapist Data

a. Assumptions

1. staffNo and patNo uniquely identify a therapist and a patient. Names are not unique.

2. A therapist works at one branch per day (given), so staffNo + appointmentDate determines branchNo.
3. A therapist sees one patient at a time, so staffNo + appointmentDate + appointmentTime identifies an appointment. patNo + appointmentDate + appointmentTime is also a candidate key; the staffNo one is chosen because the data tracks therapists.
4. A patient is not tied to a branch; the branch comes from the therapist and the day.

b. Steps 0 and 1: all the data in one table, already in 1NF

therapist_appointment: the data as given, with the appointment column split into its date and time parts.

staffNo	appointmentDate	appointmentTime	therapistName	patNo	patName	branchNo
S1011	9/12/2022	10:00	Fred Smith	P100	Lily White	M15
S1011	9/12/2022	12:00	Fred Smith	P105	Jill Baker	M15
S1024	9/12/2022	10:00	Heidi Pierce	P108	Andy McKee	Q10
S1024	9/14/2022	14:00	Heidi Pierce	P108	Andy McKee	Q10
S1032	9/14/2022	16:30	Richard Levin	P105	Jill Baker	M15
S1032	9/15/2022	18:00	Richard Levin	P110	Jimmy Winter	B13

Normal form: **1NF**. PK: staffNo, appointmentDate, appointmentTime.

Why it is already in 1NF, and the one change made:

- **appointmentDate, appointmentTime**: grouped under one "appointment" heading on the form; each gets its own column (date and time are reserved words).
- Every cell holds one value, there is no repeating group, and every row has a PK value.
- **PK**: staffNo, appointmentDate, appointmentTime. One therapist sees one patient at a time (assumption 3). staffNo is needed because two therapists both have 10:00 appointments on 9/12/2022.

The table is not in 2NF. Its dependencies:

- staffNo → therapistName (**partial**: depends on one part of the PK)
- staffNo, appointmentDate → branchNo (**partial**: a therapist works at one branch per day, whatever the time)
- staffNo, appointmentDate, appointmentTime → patNo (**full**)

- patNo → patName (**transitive**: a non-key attribute depends on another non-key attribute)

Step 2: second normal form (2NF)

Each partial dependency moves to its own relation, keyed by the part of the PK it depends on.

therapist

staffNo	therapistName
S1011	Fred Smith
S1024	Heidi Pierce
S1032	Richard Levin

therapist_schedule

staffNo	appointmentDate	branchNo
S1011	9/12/2022	M15
S1024	9/12/2022	Q10
S1024	9/14/2022	Q10
S1032	9/14/2022	M15
S1032	9/15/2022	B13

appointment

staffNo	appointmentDate	appointmentTime	patNo	patName
S1011	9/12/2022	10:00	P100	Lily White
S1011	9/12/2022	12:00	P105	Jill Baker
S1024	9/12/2022	10:00	P108	Andy McKee
S1024	9/14/2022	14:00	P108	Andy McKee
S1032	9/14/2022	16:30	P105	Jill Baker

staffNo	appointmentDate	appointmentTime	patNo	patName
S1032	9/15/2022	18:00	P110	Jimmy Winter

Why each piece changed:

- **therapistName**: moved to **therapist**, keyed by staffNo, because it depends only on staffNo.
- **branchNo**: moved to **therapist_schedule**, keyed by staffNo, appointmentDate, because a therapist works at one branch per day whatever the appointment time. The PK now enforces that rule.
- **patNo, patName**: stay in **appointment**, which keeps the full PK, because no part of the PK determines them alone.

All three relations are now in 2NF. appointment is not yet in 3NF: patNo → patName is transitive.

Step 3: third normal form (3NF)

The one transitive dependency, patNo → patName, is removed:

- **patName**: moved to a new relation, **patient**, keyed by its determinant, patNo.
- **patNo**: stays in **appointment** as an FK, so each appointment still names its patient.
- **therapist** and **therapist_schedule** have one non-key attribute each, so no non-key attribute can depend on another. They are unchanged.

The FKs that link the relations are italicized below.

therapist

staffNo	therapistName
S1011	Fred Smith
S1024	Heidi Pierce
S1032	Richard Levin

therapist_schedule

staffNo	**appointmentDate**	branchNo
S1011	9/12/2022	M15
S1024	9/12/2022	Q10
S1024	9/14/2022	Q10
S1032	9/14/2022	M15
S1032	9/15/2022	B13

patient

patNo	patName
P100	Lily White
P105	Jill Baker
P108	Andy McKee
P110	Jimmy Winter

appointment

staffNo	***appointmentDate***	**appointmentTime**	*patNo*
S1011	9/12/2022	10:00	P100
S1011	9/12/2022	12:00	P105
S1024	9/12/2022	10:00	P108
S1024	9/14/2022	14:00	P108
S1032	9/14/2022	16:30	P105
S1032	9/15/2022	18:00	P110

c/d. Final 3NF relations: keys

Relation	PK	FK
therapist	staffNo	none
therapist_schedule	staffNo, appointmentDate	staffNo → therapist.staffNo
patient	patNo	none
appointment	staffNo, appointmentDate, appointmentTime	(staffNo, appointmentDate) → therapist_schedule; patNo → patient.patNo

appointment references therapist_schedule rather than therapist, so an appointment's branch comes through that link.

Exercise 3: Maid Better Employee Hours

a. Assumptions

1. Employee names are not unique, so eName determines nothing.
2. Each event is held at one location, so eventNo determines eventLoc. Many events can share a location.
3. hours is the total an employee worked on one contract, so there is one row per employee per contract.

b. Steps 0 and 1: all the data in one table, already in 1NF

employee_hours: the data exactly as given.

eNo	contractNo	hours	eName	eventNo	eventLoc
1135	C1024	16	Smith J	H25	Queens
1057	C1024	24	Hocine D	H25	Queens
1068	C1025	28	White T	H4	Yonkers
1135	C1025	15	Smith J	H4	Yonkers
1135	C1026	10	Smith J	H25	Queens

Normal form: **1NF**. PK: eNo, contractNo.

Why it is already in 1NF:

- Every cell holds one value and there is no repeating group.
- **PK**: eNo, contractNo. Neither is unique alone (1135, C1024 and C1025 repeat), but together they are (assumption 3).

The table is not in 2NF. Its dependencies:

- eNo, contractNo → hours (**full**)
- eNo → eName (**partial**: depends on one part of the PK)
- contractNo → eventNo, eventLoc (**partial**)
- eventNo → eventLoc (**transitive**: a non-key attribute depends on another non-key attribute)

Step 2: second normal form (2NF)

Each partial dependency moves to its own relation, keyed by the part of the PK it depends on.

employee

eNo	eName
1135	Smith J
1057	Hocine D
1068	White T

contract

contractNo	eventNo	eventLoc
C1024	H25	Queens
C1025	H4	Yonkers
C1026	H25	Queens

employee_contract

eNo	contractNo	hours
1135	C1024	16
1057	C1024	24
1068	C1025	28
1135	C1025	15
1135	C1026	10

Why each piece changed:

- **eName**: moved to **employee**, keyed by eNo, because it depends only on eNo.
- **eventNo, eventLoc**: moved to **contract**, keyed by contractNo, because they depend only on contractNo.
- **hours**: stays with the full PK in the remaining relation, named **employee_contract**, because it depends on both the employee and the contract.

All three relations are now in 2NF. contract is not yet in 3NF: eventNo → eventLoc is transitive.

Step 3: third normal form (3NF)

The one transitive dependency, eventNo → eventLoc, is removed:

- **eventLoc**: moved to a new relation, **event**, keyed by its determinant, eventNo.
- **eventNo**: stays in **contract** as an FK, so each contract still names its event.
- **employee** and **employee_contract** have one non-key attribute each, so no non-key attribute can depend on another. They are unchanged.

The FKs that link the relations are italicized below.

employee

eNo	eName
1135	Smith J
1057	Hocine D
1068	White T

event

eventNo	eventLoc
H25	Queens
H4	Yonkers

contract

contractNo	*eventNo*
C1024	H25
C1025	H4
C1026	H25

employee_contract

eNo	***contractNo***	hours
1135	C1024	16
1057	C1024	24
1068	C1025	28
1135	C1025	15
1135	C1026	10

c/d. Final 3NF relations: keys

Relation	PK	FK
employee	eNo	none
event	eventNo	none
contract	contractNo	eventNo → event.eventNo

Relation	PK	FK
employee_contract	eNo, contractNo	eNo → employee.eNo; contractNo → contract.contractNo